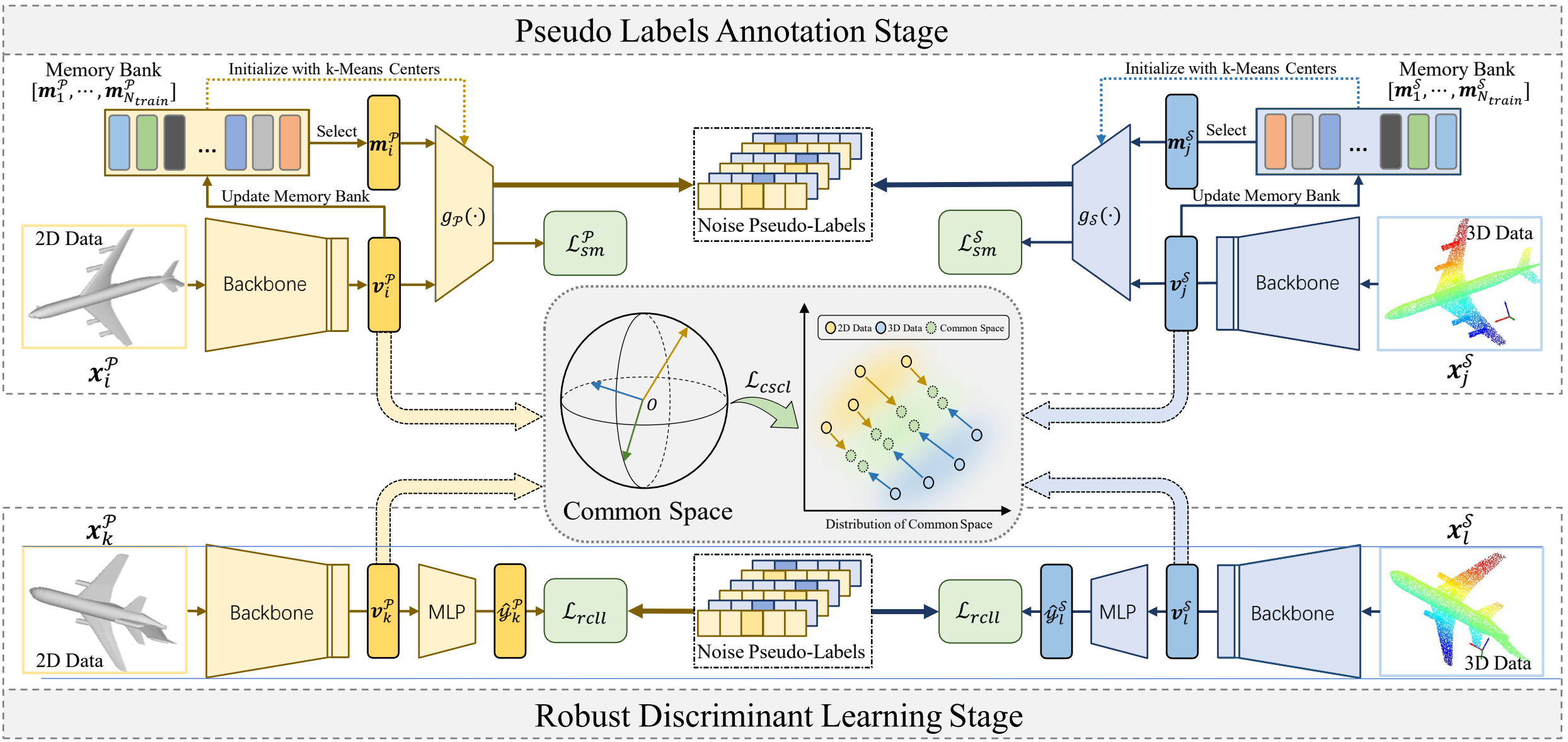
Pseudo Labels Annotation Stage
Memory Bank
$[\boldsymbol{m}_1^{\mathcal{P}}, \cdots, \boldsymbol{m}_{N_{train}}^{\mathcal{P}}]$
Initialize with k-Means Centers
Select
$\boldsymbol{m}_i^{\mathcal{P}}$
Update Memory Bank
$g_{\mathcal{P}}(\cdot)$
$\mathcal{L}_{sm}^{\mathcal{P}}$
Noise Pseudo-Labels
$\mathcal{L}_{sm}^{\mathcal{S}}$
$g_{\mathcal{S}}(\cdot)$
$\boldsymbol{m}_j^{\mathcal{S}}$
Select
Initialize with k-Means Centers
Memory Bank
$[\boldsymbol{m}_1^{\mathcal{S}}, \cdots, \boldsymbol{m}_{N_{train}}^{\mathcal{S}}]$
Update Memory Bank
2D Data
Backbone
$\boldsymbol{v}_i^{\mathcal{P}}$
$\boldsymbol{x}_i^{\mathcal{P}}$
$\boldsymbol{v}_j^{\mathcal{S}}$
Backbone
3D Data
$\boldsymbol{x}_j^{\mathcal{S}}$
O
$\mathcal{L}_{cscl}$
2D Data
3D Data
Common Space
Common Space
Distribution of Common Space
$\boldsymbol{x}_k^{\mathcal{P}}$
2D Data
Backbone
$\boldsymbol{v}_k^{\mathcal{P}}$
MLP
$\hat{y}_k^{\mathcal{P}}$
$\mathcal{L}_{rcll}$
Noise Pseudo-Labels
$\mathcal{L}_{rcll}$
$\hat{y}_l^{\mathcal{S}}$
MLP
$\boldsymbol{v}_l^{\mathcal{S}}$
Backbone
$\boldsymbol{x}_l^{\mathcal{S}}$
3D Data
Robust Discriminant Learning Stage